BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Paper 1: Networks and Security

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. How do you calculate raw text size in bits?

2. What can a subprogram's parameter receive?
   A) Only integers
   B) An array
   C) Only strings
   D) Only Boolean values

3. "readLine() writes a new line into the file." Correct this. Explain why.

4. A program saves a result to a text file. "writeLine(text) only shows text on screen." Correct this. How should the program save the result?

Today's objectives

Explain the core ideas of paper 1: networks and security using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Network choice
Distinguish LAN/WAN, client-server/peer-to-peer and suitable devices.

Internet services
DNS maps domain names to IP addresses; routers connect networks.

Recognise attacks
Distinguish phishing, malware and denial of service.

Prevent attacks
Combine least privilege, authentication, encryption, updates and backups.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Network choice
Distinguish LAN/WAN, client-server/peer-to-peer and suitable devices.

Internet services
DNS maps domain names to IP addresses; routers connect networks.

Recognise attacks
Distinguish phishing, malware and denial of service.

Prevent attacks
Combine least privilege, authentication, encryption, updates and backups.

I Do - Worked model

A school receives a phishing email directing staff to an unfamiliar login page. Recommend two defences.

- Identify credential theft/social engineering.
- Train staff to verify suspicious links through another channel.
- Use MFA so a password alone is insufficient.
- Explain both measures target different stages of compromise.

We Do - Guided practice

1. A device has poor Wi-Fi but fast wired access. Explain two possible causes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the comparison so a warning appears on the third failed attempt.

```
failed = 3
if failed > 3 then
print("Review access")
endif
```

You Do - Independent exam questions

Q2. State one role of DNS. [2 marks]

Q3. Name the device that connects networks. [1 mark]

Q4. Explain why encryption alone does not prevent phishing. [3 marks]

Exit Ticket - True or False?

Encryption stops all social engineering attacks.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Number of characters x bits per character, using the given encoding.

2. B - An array

3. It reads the next line and advances the current position.

4. writeLine(text) writes a line through an open file handle. Open the file, write, then close it. Printing on screen does not save it.

I Do - Explain the model

Identify credential theft/social engineering.

Train staff to verify suspicious links through another channel.

Use MFA so a password alone is insufficient.

Explain both measures target different stages of compromise.

We Do - Model answers

1. Do not assume internet service itself must be slow.

Code First - Answer

```
failed = 3
if failed >= 3 then
print("Review access")
endif
```

Expected output: Review access

You Do - Self-mark

Q2: Resolves domain names to IP addresses.

Q3: Router.

Q4: Encryption protects data in transit but cannot stop a user willingly entering credentials into a fraudulent site.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.